

$$\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \text{ is a vector}$$

→ n dimensional but we
- vectors

ψ wave function can be a vector so? |ψ⟩
∇ψ ∇·ψ ∇×ψ are all valid



$$\begin{aligned} \mathcal{L}_{SM} = & -\frac{1}{2}\partial_\mu g_\mu^a g_\mu^a - g_s f^{abc}\partial_\mu g_\mu^a g_\mu^b g_\mu^c - \frac{1}{2}g_s^2 f^{abc}f^{ade}g_\mu^b g_\mu^c g_\mu^d g_\mu^e - \partial_\mu W_\mu^+ \partial_\mu W_\mu^- \\ & - M^2 W_\mu^+ W_\mu^- - \frac{1}{2}\partial_\mu Z_\mu^0 \partial_\mu Z_\mu^0 - \frac{1}{2}M_Z^2 Z_\mu^0 Z_\mu^0 - \frac{1}{2}\partial_\mu A_\mu \partial_\mu A_\mu - igc_w(\partial_\mu Z_\mu^0(W_\mu^+ W_\mu^- - \\ & W_\mu^- W_\mu^+) - Z_\mu^0(W_\mu^+ \partial_\mu W_\mu^- - W_\mu^- \partial_\mu W_\mu^+) + Z_\mu^0(W_\mu^+ \partial_\mu W_\mu^- - W_\mu^- \partial_\mu W_\mu^+)) - \\ & ig s_w(\partial_\mu A_\mu(W_\mu^+ W_\mu^- - W_\mu^- W_\mu^+) - A_\mu(W_\mu^+ \partial_\mu W_\mu^- - W_\mu^- \partial_\mu W_\mu^+) + A_\mu(W_\mu^+ \partial_\mu W_\mu^- - \\ & W_\mu^- \partial_\mu W_\mu^+)) - \frac{1}{2}g^2 W_\mu^+ W_\mu^- W_\mu^+ W_\mu^- + \frac{1}{2}g^2 W_\mu^+ W_\mu^- W_\mu^+ W_\mu^- + g^2 c_w^2(Z_\mu^0 W_\mu^+ Z_\mu^0 W_\mu^- - \\ & Z_\mu^0 Z_\mu^0 W_\mu^+ W_\mu^-) + g^2 s_w^2(A_\mu W_\mu^+ A_\mu W_\mu^- - A_\mu A_\mu W_\mu^+ W_\mu^-) + g^2 s_w c_w(A_\mu Z_\mu^0(W_\mu^+ W_\mu^- - \\ & W_\mu^- W_\mu^+) - 2A_\mu Z_\mu^0 W_\mu^+ W_\mu^-) - \frac{1}{2}\partial_\mu H \partial_\mu H - 2M^2 \alpha_h H^2 - \partial_\mu \phi^+ \partial_\mu \phi^- - \frac{1}{2}\partial_\mu \phi^0 \partial_\mu \phi^0 - \\ & \beta_h \left(\frac{2M_h^2}{g^2} + \frac{2M_h^2}{g^2} H + \frac{1}{2}(H^2 + \phi^0 \phi^0 + 2\phi^+ \phi^-) \right) + \frac{2M_h^2}{g^2} \alpha_h - \\ & \frac{g\alpha_h M}{2} (H^3 + H\phi^0 \phi^0 + 2H\phi^+ \phi^-) - \\ & \frac{1}{8}g^2 \alpha_h (H^4 + (\phi^0)^4 + 4(\phi^+ \phi^-)^2 + 4(\phi^0)^2 \phi^+ \phi^- + 4H^2 \phi^+ \phi^- + 2(\phi^0)^2 H^2) - \\ & g M W_\mu^+ W_\mu^- H - \frac{1}{2}g \frac{M}{c_w} Z_\mu^0 Z_\mu^0 H - \\ & \frac{1}{2}ig (W_\mu^+ (\phi^0 \partial_\mu \phi^- - \phi^- \partial_\mu \phi^0) - W_\mu^- (\phi^0 \partial_\mu \phi^+ - \phi^+ \partial_\mu \phi^0)) + \\ & \frac{1}{2}g (W_\mu^+ (H \partial_\mu \phi^- - \phi^- \partial_\mu H) + W_\mu^- (H \partial_\mu \phi^+ - \phi^+ \partial_\mu H)) + \frac{1}{2}g \frac{1}{c_w} (Z_\mu^0 (H \partial_\mu \phi^0 - \phi^0 \partial_\mu H) + \\ & M (\frac{1}{c_w} Z_\mu^0 \partial_\mu \phi^0 + W_\mu^+ \partial_\mu \phi^- + W_\mu^- \partial_\mu \phi^+) - ig \frac{M}{c_w} Z_\mu^0 (W_\mu^+ \phi^- - W_\mu^- \phi^+) + ig s_w M A_\mu (W_\mu^+ \phi^- - \\ & W_\mu^- \phi^+) - ig \frac{1-2s_w^2}{2c_w} Z_\mu^0 (\phi^+ \partial_\mu \phi^- - \phi^- \partial_\mu \phi^+) + ig s_w A_\mu (\phi^+ \partial_\mu \phi^- - \phi^- \partial_\mu \phi^+) - \\ & \frac{1}{4}g^2 W_\mu^+ W_\mu^- (H^2 + (\phi^0)^2 + 2\phi^+ \phi^-) - \frac{1}{8}g^2 \frac{1}{c_w} Z_\mu^0 Z_\mu^0 (H^2 + (\phi^0)^2 + 2(2s_w^2 - 1)^2 \phi^+ \phi^-) - \\ & \frac{1}{2}g^2 \frac{1}{c_w} Z_\mu^0 \phi^0 (W_\mu^+ \phi^- + W_\mu^- \phi^+) - \frac{1}{2}ig^2 \frac{1}{c_w} Z_\mu^0 H (W_\mu^+ \phi^- - W_\mu^- \phi^+) + \frac{1}{2}g^2 s_w A_\mu \phi^0 (W_\mu^+ \phi^- + \\ & W_\mu^- \phi^+) + \frac{1}{2}ig^2 s_w A_\mu H (W_\mu^+ \phi^- - W_\mu^- \phi^+) - g^2 \frac{1-2s_w^2}{2c_w} (2c_w^2 - 1) Z_\mu^0 A_\mu \phi^+ \phi^- - \\ & g^2 s_w^2 A_\mu A_\mu \phi^+ \phi^- + \frac{1}{2}ig_s \lambda_\phi^2 (\bar{q} \gamma^\mu q) g_\mu^a - e^2 (\gamma \partial + m_\phi^2) e^\lambda - \bar{\nu}^\lambda (\gamma \partial + m_\nu^2) \nu^\lambda - \bar{u}_j^\lambda (\gamma \partial + \\ & m_u^2) u_j^\lambda - \bar{d}_j^\lambda (\gamma \partial + m_d^2) d_j^\lambda + ig s_w A_\mu (-e^\lambda \gamma^\mu e^\lambda + \frac{2}{3}(\bar{u}_j^\lambda \gamma^\mu u_j^\lambda - \frac{1}{3}(\bar{d}_j^\lambda \gamma^\mu d_j^\lambda)) + \\ & \frac{1}{6}c_w Z_\mu^0 \{(\bar{\nu}^\lambda \gamma^\mu (1 + \gamma^5) \nu^\lambda) + (e^\lambda \gamma^\mu (4s_w^2 - 1 - \gamma^5) e^\lambda) + (\bar{d}_j^\lambda \gamma^\mu (\frac{2}{3}s_w^2 - 1 - \gamma^5) d_j^\lambda) + \\ & (\bar{u}_j^\lambda \gamma^\mu (1 - \frac{2}{3}s_w^2 + \gamma^5) u_j^\lambda)\} + \frac{1}{2\sqrt{2}} W_\mu^+ ((\bar{\nu}^\lambda \gamma^\mu (1 + \gamma^5) U^{lep}{}_{\lambda e}) + (\bar{u}_j^\lambda \gamma^\mu (1 + \gamma^5) C_{\lambda e} d_j^\lambda)) + \\ & \frac{1}{2\sqrt{2}} W_\mu^- ((e^\lambda U^{lep}{}_{e\lambda} \gamma^\mu (1 + \gamma^5) \nu^\lambda) + (\bar{d}_j^\lambda C_{\lambda e}^\dagger \gamma^\mu (1 + \gamma^5) u_j^\lambda)) + \\ & \frac{1}{2M\sqrt{2}} \phi^+ (-m_\nu^2 (\bar{\nu}^\lambda U^{lep}{}_{\lambda e} (1 - \gamma^5) e^\lambda) + m_\nu^2 (\bar{\nu}^\lambda U^{lep}{}_{\lambda e} (1 + \gamma^5) e^\lambda) + \\ & \frac{1}{2M\sqrt{2}} \phi^- (m_\nu^2 (e^\lambda U^{lep}{}_{\lambda e} (1 + \gamma^5) \nu^\lambda) - m_\nu^2 (e^\lambda U^{lep}{}_{\lambda e} (1 - \gamma^5) \nu^\lambda) - \frac{g}{2} \frac{m_\nu^2}{M} H (\bar{\nu}^\lambda \nu^\lambda) - \\ & \frac{g}{2} \frac{m_\nu^2}{M} H (e^\lambda e^\lambda) + \frac{1}{2} \frac{m_\nu^2}{M} \phi^0 (\bar{\nu}^\lambda \gamma^5 \nu^\lambda) - \frac{1}{2} \frac{m_\nu^2}{M} \phi^0 (e^\lambda \gamma^5 e^\lambda) - \frac{1}{4} \bar{\nu}_\lambda M_\lambda^R (1 - \gamma_5) \bar{\nu}_\lambda - \\ & \frac{1}{4} \bar{\nu}_\lambda M_\lambda^R (1 - \gamma_5) \bar{\nu}_\lambda + \frac{1}{2M\sqrt{2}} \phi^+ (-m_\nu^2 (\bar{u}_j^\lambda C_{\lambda e} (1 - \gamma^5) d_j^\lambda) + m_\nu^2 (\bar{u}_j^\lambda C_{\lambda e} (1 + \gamma^5) d_j^\lambda) + \\ & \frac{1}{2M\sqrt{2}} \phi^- (m_\nu^2 (\bar{d}_j^\lambda C_{\lambda e}^\dagger (1 + \gamma^5) u_j^\lambda) - m_\nu^2 (\bar{d}_j^\lambda C_{\lambda e}^\dagger (1 - \gamma^5) u_j^\lambda) - \frac{g}{2} \frac{m_\nu^2}{M} H (\bar{u}_j^\lambda u_j^\lambda) - \\ & \frac{g}{2} \frac{m_\nu^2}{M} H (\bar{d}_j^\lambda d_j^\lambda) + \frac{1}{2} \frac{m_\nu^2}{M} \phi^0 (\bar{u}_j^\lambda \gamma^5 u_j^\lambda) - \frac{1}{2} \frac{m_\nu^2}{M} \phi^0 (\bar{d}_j^\lambda \gamma^5 d_j^\lambda) + \bar{G}^a \partial^2 G^a + g_s f^{abc} \partial_\mu \bar{G}^a G^b g_\mu^c + \\ & \bar{X}^+ (\partial^2 - M^2) X^+ + \bar{X}^- (\partial^2 - M^2) X^- + \bar{X}^0 (\partial^2 - \frac{M^2}{2}) X^0 + \bar{Y} \partial^2 Y + ig c_w W_\mu^+ (\partial_\mu \bar{X}^0 X^- - \\ & \partial_\mu \bar{X}^- X^+) + ig s_w W_\mu^+ (\partial_\mu \bar{X}^- X^- - \partial_\mu \bar{X}^+ X^+) + ig c_w W_\mu^- (\partial_\mu \bar{X}^- X^0 - \\ & \partial_\mu \bar{X}^0 X^+) + ig s_w W_\mu^- (\partial_\mu \bar{X}^- X^- - \partial_\mu \bar{X}^+ X^+) + ig c_w Z_\mu^0 (\partial_\mu \bar{X}^+ X^+ - \\ & \partial_\mu \bar{X}^- X^-) + ig s_w A_\mu (\partial_\mu \bar{X}^+ X^+ - \\ & \partial_\mu \bar{X}^- X^-) - \frac{1}{2}g M (\bar{X}^+ X^+ H + \bar{X}^- X^- H + \frac{1}{2} \bar{X}^0 X^0 H) + \frac{1-2s_w^2}{2c_w} ig M (\bar{X}^+ X^0 \phi^+ - \bar{X}^- X^0 \phi^-) + \\ & \frac{1}{2c_w} ig M (\bar{X}^0 X^- \phi^+ - \bar{X}^0 X^+ \phi^-) + ig M s_w (\bar{X}^0 X^- \phi^+ - \bar{X}^0 X^+ \phi^-) + \\ & \frac{1}{2}ig M (\bar{X}^+ X^+ \phi^0 - \bar{X}^- X^- \phi^0) . \end{aligned}$$